



Alexa on Amazon App

Ask Alexa about news, weather, basic facts, or use the assistant's add-on "skills," right from the Amazon app. <http://dgit.in/AlexaAm>



Get paid to drive

A company by the name of Payver is working on Lvl5 and will pay you a few pennies for recording the road when you drive. <http://dgit.in/Payver5>

Sunny side up

While the sun has been powering us since time immemorial, we've truly begun harnessing its power quite recently.

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olar energy has been touted as the answer to rising energy woes around the world for what seems like ages

now. But it is only recently that we have seen a significant effort from various countries by harnessing the power of our local star. And it's about time because life on our planet is itself powered by the sun. The food chain, that we are so proud of being on the top of, exists due to the sunlight harnessed by chlorophyll in plants. In fact, practically everything around you – ranging from the fossil fuel guzzling machines to essential natural phenomenon like the water cycle – happens because of the sun's energy. And fortunately, it isn't running out or falling short anytime soon.

According to the United Nations Development Programme estimates, the total annual solar potential of earth ranges between 1,575–49,837 exajoules (EJ), where even the lower estimate is more than twice the annual consumption of the planet – around 500 exajoules. Interestingly, that is just 0.0000013th of the total energy produced by the Sun each second. To be honest, that does not include the 3000 exajoules that plants consume by photosynthesis annually. Sadly, we hardly use any of this for generating electricity, yet.

Unlike what some world leaders would have you believe, our own country is also significantly invested in the future of solar energy. The Kurnool Ultra Mega Solar Park and the Kamuthi Solar Project are two of the world's largest solar power projects

with a combined capacity of more than 1500MW. In fact, India and China combined have the five largest solar projects currently in the world. Although, getting here wasn't easy. Solar energy, or rather the interest in using it as a direct source of power, reportedly go way back to the 7th century BC. But actual literary evidence of it being used comes much later.

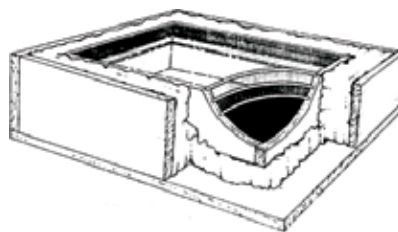
ANCIENT TIMES

In second century BC, Greek scientist Archimedes reportedly used giant reflective bronze shields to focus sunlight onto incoming Roman ships, in order to set them ablaze. While no concrete evidence of this feat exists, the Greek navy repeated this experiment in 1973 and were successfully able to set a wooden boat ablaze at a distance of 50 metres. But even before that people have been using reflected sunlight to illuminate corridors and doorways, especially in Egyptian tombs. This also enabled them to trap the heat from the

sunlight for lower daytime and higher evening temperatures.

Apart from this, energy concentrated by reflecting sunlight was also associated with a religious significance, mostly because most civilisations considered the sun as a religious entity. Eventually, the benefits were pretty evident – so much so, that the Romans mandated a sun room in every house by law before 600 AD in the Justinian code.

Taking energy from the heat of sunlight was later applied to more extensive concepts like the solar oven. Swiss inventor Horace de Saussure made the first Solar Collector in 1767. The design consisted of an insulated box with a glass top and a couple of smaller boxes inside. In the sun, the internal boxes reached temperatures around 100 degrees Celsius quite easily. This design was later improved in the 1800s by Sir John Herschel by introducing a black inner-lining and sand into the design. While work in this area continued until the first world war, with innovations like the parabolic trough, it was the advent of photovoltaic cells that changed solar energy forever.



A representative image for an improved Saussure hot box. To make your own DIY solar cooker, check out <http://dgit.in/SolrDIY>

PHOTOVOLTAIC

The exact invention of the solar panel is under debate because of the number of iterative contributions it took for a modern practical solar design to come around. Some associate the invention to French physicist Edmond Becquerel